LLM Fine-Tuning Frameworks

Frameworks, comparison matrix, enterprise expectations and key research papers

1. LLM Fine-Tuning Frameworks

Important Resources (LLM Leaderboard)

- https://github.com/pdaicode/awesome-LLMs-finetuning
- https://github.com/Curated-Awesome-Lists/awesome-llms-fine-tuning/tree/main
- https://github.com/mamur/awesome-finetuning

Hugging Face

- Huge Model Hub (300,000+ models)
- Transformers Library = Industry Standard
- Fine-Tuning Made Easy
- PEFT + LoRA for Efficient Fine-Tuning
- Datasets Library
- Trainer + Accelerate = Fast & Distributed
- Hugging Face Hub
- Evaluation, Tokenizers, & Inference Support

DeepSpeed

- Memory Optimization (ZeRO)
- Faster Training (FP16 / BF16, Offloading, Kernel Fusion)
- Gradient Accumulation + Mixed Precision
- Model Parallelism = Pipeline Parallelism
- Checkpointing + Resume + Logging

LLaMA Factory

- LoRA / QLoRA Fine-Tuning
- Intuitive Setup, GUI-friendly
- Supports 50+ Hugging Face LLMs
- Multi-GPU, DeepSpeed, FSDP
- Chat Templates + SFT Format Support
- Integrated Evaluation + MMLU / TruthfulQA
- Export to GGUF / GPTQ / HF

Unsloth

- Fastest LoRA Fine-Tuning
- Low VRAM Usage — Train LLM on Colab T4
- One Line LLM Loading
- Integrated PEFT + Hugging Face Compatibility
- Supports Multiple Model and Multimodal
- Torch Compile + Flash Attention

Axolotl

- All-in-One Fine-Tuning Powerhouse
- No Code — Just YAML Config
- Supports All Open LLMs (100+ HF models)
- DeepSpeed + FSDP + Flash Attention Support
- Smart Prompt Templates + Chat Datasets
- Post-Finetune Export
- Active Dev Community

Colossal-AI

- Multiple Parallelism Techniques
- Zero Redundancy Optimizer (ZeRO)
- Auto-Parallelism with Colossal-Auto
- Memory Optimization
- Integration with PyTorch

LightLLM / vLLM

- Paged Attention (Memory-efficient attention)
- Batching engine
- Supports HF models
- OpenAI-Compatible API
- Streaming support
- Quantization compatible

OpenLLM (BentoML)

- Production-Ready LLM Serving
- Supports Fine-Tuned Models from HF + PEFT
- BentoML Ecosystem = Deployment Superpowers
- Multi-Model Serving
- Auto-Generate API + CLI
- Easy Integration with LangChain, LlamaIndex

FastChat (LMSYS)

- Fine-Tuning Support for Open LLMs
- Chat-Based Instruction Fine-Tuning
- Multi-GPU + DeepSpeed Training
- Fast Inference & Serving
- OpenAI-Compatible API

SkyPilot

- Multi-Cloud Support (AWS, GCP, Azure, Lambda Cloud, etc.)
- GPU Spot Auto-Recovery
- Checkpoint Mounting
- Cluster Abstraction
- Auto-Tuner
- Job Scheduling
- Easy Integration

2. Framework Comparison Table

Ratings are conservative and aligned with the notes' goal descriptions and recommendations. Frameworks are not all-in-one tools; infrastructure and serving tools receive partial marks for training goals.

Framework	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
Hugging Face (Transformers + PEFT + TRL)	BEST	YES	BEST	YES	~	YES	YES	~	~	~	~	~
LLaMA Factory	YES	BEST	~	YES	~	~	BEST	BEST	~	~	~	~
Axolotl	BEST	YES	BEST	BEST	~	YES	BEST	~	YES	YES	~	~
Unsloth	YES	BEST	~	~	~	~	~	BEST	~	~	~	~
SkyPilot	~	~	~	YES	~	~	~	~	YES	BEST	YES	BEST
DeepSpeed	YES	YES	~	BEST	~	~	~	~	BEST	YES	~	~
FSDP	YES	YES	~	BEST	~	~	~	~	YES	YES	~	~
Colossal-AI	YES	YES	~	BEST	~	~	~	~	BEST	BEST	~	YES
OpenLLM	~	~	~	~	BEST	YES	~	~	~	~	YES	BEST
FastChat	YES	~	~	YES	BEST	BEST	~	~	~	~	BEST	YES
vLLM	~	YES	~	~	BEST	BEST	~	~	~	~	BEST	YES
LightLLM	~	YES	~	~	~	~	~	~	~	~	BEST	~

Legend		
	BEST	Industry-best for that goal (based on the notes' recommendations)
	YES	Fully supported
	~	Possible, but requires effort or is not the primary goal

3. Fine-Tuning Enterprise-Level Expectations

When a company plans to fine-tune LLMs at the enterprise level, their expectations are not limited to just training a model. They look for an entire structured pipeline, proper governance, cost optimization, and planning that is centered around the actual use case.

Process & Governance

- Clear Use-Case Definition
- High-Quality, Domain-Specific Dataset
- Data Governance & Privacy
- Model Selection & Fine-Tuning Strategy (Base Model Decision; Fine-Tuning Type)
- Infrastructure & Compute Requirements
- Fine-tuned models lifecycle managed (MLOps & CI/CD for Models), Deployment
- Evaluation & Benchmarking
- Risk Management & Guardrails
- Cost & ROI Awareness
- Explainability & Auditing

Framework Qualities

- Fast Training = Fast Time-to-Market
- Adaptive Framework = Plug-n-Play Architecture
- Dynamic Resource Adaptation
- Auto-Resume & Fault Tolerance
- Minimal Code Change, Maximum Flexibility
- Community Support

Smart framework choice: Hugging Face + PEFT / LLaMA-Factory / Unsloth / DeepSpeed / Colossal-AI / Axolotl

4. Important Research Papers

Model / Paper	Year	Training Strategy	Highlights
BERT	2018	Masked Language Modeling (MLM) + Next Sentence Prediction (NSP)	Bidirectional encoder; revolutionized NLP tasks like QA and NER.
GPT	2018	Autoregressive Language Modeling + Supervised Fine-Tuning	Unidirectional decoder; laid the foundation for ChatGPT-style models.
ULMFiT	2018	LM pretraining, then task-specific fine-tuning with discriminative learning rates	First practical transfer learning approach in NLP; enabled effective fine-tuning on small datasets.
T5	2020	Unified Text-to-Text Transfer Transformer	Converted all NLP tasks into text-to-text format; pretrained on C4 dataset.
LLaMA 2	2023	Pretrained on 2T tokens; fine-tuned for dialogue (LLaMA-2-Chat)	Open-source models ranging from 7B to 70B parameters; optimized for dialogue use cases.
Mistral 7B	2023	Grouped-Query Attention (GQA) and Sliding Window Attention (SWA)	Efficient 7B model; optimized for reasoning and code generation.
DeepSeek-R1	2024	Reinforcement Learning without prior supervised fine-tuning	Demonstrates strong reasoning capabilities; trained via large-scale RL.
GPT-3	2020	Demonstrated prompting over fine-tuning	Prompting vs. fine-tuning philosophy; gave rise to RAG and agents.

Mixture-of-Experts (MoE)

Sparse Mixture-of-Experts architecture with expert routing. Scales model capacity efficiently; only a subset of experts is activated per input, reducing computation.

Model	Creator / Org	Details
Switch Transformer	Google	First scalable MoE model; uses 1 active expert per token; 100B+ parameters
GShard	Google Brain	Predecessor to Switch Transformer; introduced MoE scaling for multilingual
GLaM	Google	1.2T parameters but uses only 97B per token — highly efficient
Mixtral	Mistral AI	Open-weight MoE model, 12.9B parameters total (2 of 8 experts active)
DeepSpeed-MoE	Microsoft	MoE toolkit for building sparse LLMs using DeepSpeed

5. Research Papers with Official Links

LoRA (2021) — Low-Rank Adaptation
Concept: Trains low-rank matrices injected into Transformer layers for efficient fine-tuning.
Paper: https://arxiv.org/abs/2106.09685

Adapters (2019) — Parameter-Efficient Transfer Learning for NLP
Concept: Inserts small trainable adapter modules into frozen models.
Paper: https://arxiv.org/abs/1902.00751

Prefix-Tuning (2021) — Optimizing Continuous Prompts for Generation
Concept: Optimizes a small continuous task-specific vector (prefix) while keeping model parameters frozen.
Paper: https://arxiv.org/abs/2101.00190

QLoRA (2023) — Efficient Fine-Tuning of Quantized LLMs
Concept: Combines 4-bit quantization with LoRA for memory-efficient fine-tuning.
Paper: https://arxiv.org/abs/2305.14314

InstructGPT (2022) — Fine-Tuning LMs from Human Preferences
Concept: Combines supervised fine-tuning with reinforcement learning from human feedback (RLHF).
Paper: https://arxiv.org/abs/2203.02155

DPO (2023) — Direct Preference Optimization
Concept: Simplifies preference optimization by directly optimizing the policy without a separate reward model.
Paper: https://arxiv.org/abs/2305.18290

Self-Instruct (2022)
Concept: Models generate their own instructions to fine-tune without human labels.
Paper: https://arxiv.org/abs/2212.10560

Scaling Laws for Neural Language Models (2020)
Concept: Demonstrates how model performance scales with model size, data, and compute.
Paper: https://arxiv.org/abs/2001.08361

Efficient Training to Fill in the Middle (2022)
Concept: Introduces a strategy to fine-tune models on fill-in-the-middle tasks.
Paper: https://arxiv.org/abs/2207.14255